Which process faults can be told apart? Diagnosability limits of in-situ monitoring in laser powder bed fusion

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Abstract

In-situ monitoring is standard on metal additive manufacturing systems, and classifiers report high accuracies for detecting defects from photodiode and imaging signals. Those results answer a question about outcomes: which defect formed. Specifying hardware requires an answer one step upstream — which process *causes* a sensor configuration can separate in principle, and which remain confounded whatever classifier is applied. Where two causes produce signatures differing by less than the sensor noise, no volume of training data distinguishes them; the limit is structural, not algorithmic. This paper applies structural and quantitative diagnosability analysis, mature in automotive and aerospace process control, to laser powder bed fusion. A calibrated reduced-order thermal model generates the response of a coaxial suite to five injected faults: laser power and scan speed deviation, thin powder layers from recoater short-feed, degraded shielding gas flow, and geometry-driven heat accumulation, with separability measured as an angle between noise-whitened fault signatures. Laser power and gas degradation prove *exactly* confounded instantaneously, by algebra rather than noise, so no instrument reading the melt pool separates them, and seven candidate observables raise the attainable rank only from three to four. Regressing the same three sensors onto known build context raises the rank to five and the worst pair from 0.00° to 25.83° at no hardware cost, outperforming two idealised direct measurements of the causes themselves. Diagnosability matrices are reported for five configuration classes typical of commercial machines.

Keywords: Additive manufacturing, Process monitoring, Diagnosability, Fault diagnosis, Laser powder bed fusion

1. Introduction

Laser powder bed fusion (LPBF) builds a part by melting thousands of short track segments per layer, and a modest excursion in any of several independent inputs can produce a rejected part. Machine manufacturers have responded by fitting in-situ monitoring as standard — coaxial photodiodes, melt pool cameras, pyrometers, and increas-

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ingly acoustic and powder-bed imaging — and a large literature maps those signals onto defects with high reported accuracy.

Almost all of that work answers a question about *outcomes*: given the signals, which defect formed? That matters for part acceptance, but it is not the question a machine builder asks when specifying hardware, nor the one a process engineer asks when a build drifts. Those concern *causes* — the laser is under-delivering, the recoater is short-feeding, the gas flow has degraded — and which it is determines what gets corrected.

The distinction is not semantic, because causes can be mutually indistinguishable in a way outcomes are not. If two causes drive the readings along the same direction in signal space, the readings contain no information separating them, and no classifier, however trained, can recover the difference. The limitation is structural — a property of the measurement and the physics, not of the algorithm. Conversely, a pair separable in principle but misclassified in practice is an algorithm that can be improved.

Diagnosability analysis is the established discipline for deciding which case holds. Developed within model-based fault detection and isolation (FDI), it asks, before any estimator is built, which faults a given measurement set can detect and which it can isolate from one another [1, 2]. The methods are standard where sensor sets are costly and must be justified in advance, but to the authors' knowledge they have not been applied to LPBF.

This paper applies them. A calibrated reduced-order thermal model is driven by five injected fault modes and the resulting signatures are compared pairwise against noise, yielding a diagnosability matrix per configuration that divides fault pairs into those resolvable from existing signals, those needing an added measurement, and those structurally irreducible. The contributions are the first such matrix for the process, to the authors' knowledge; a demonstration that laser power and shielding gas degradation are *exactly* confounded instantaneously, so no melt-pool instrument separates them; the finding that the binding limit is the melt pool rather than the instrumentation; and that build structure already present dissolves that limit at no hardware cost, outperforming two idealised direct measurements of the causes.

2. Background and related work

2.1. What in-situ monitoring classifies

Melt pool monitoring with coaxial photodiodes and cameras is the most widely deployed configuration, and its canonical application is flaw classification. Scime and Beuth identified signatures indicative of keyholing porosity and balling from high-speed visible imaging [3], and later work extended this across modalities and alloys, including mechanical-property prediction from coaxial signals [4]. The predicted quantity is in each case an outcome or a material state, and whether two distinct excursions could have produced the same signal is not posed.

A smaller body of work estimates process inputs, and is closer to the present question. RAISE-LPBF reconstructs laser power and scan speed independently from on-axis video on 316L, sampling both per scan line from continuous distributions so interactions can be studied [5]. Reijonen reports coaxial photodiode response to power,

speed, focal position, scan orientation and shielding gas flow, finding power approximately linear, speed a power law, and the remainder non-linear and interactive [6]. These characterise *sensitivity*; identifiability — whether sensitivities to different causes are linearly distinguishable — is the question neither addresses.

Sensing that observes the powder bed forms a third class, attacking causes the melt pool reports only indirectly: Liu et al. recover local powder layer thickness and thermal transport from the post-recoat heat-up transient [8], and Bugatti and Colosimo detect geometric and surface anomalies below 100 µm with a recoater-mounted contact image sensor [9]. Both need added hardware; neither is simulated here, and they appear in Section 4.7 as targets rather than modelled options.

2.2. Lumped process metrics are incomplete

A recurring pattern is to report a *lumped* process quantity in place of its components. Volumetric energy density (VED) is the usual choice, and state-of-the-art parameter estimation from melt pool imagery still predicts energy density *class* rather than power and speed separately [10].

There is direct evidence that this discards information. Guo et al. held input energy density constant while varying power and speed together, and observed melt pool dimensions and volume changing by orders of magnitude with transitions between melting regimes, attributing it to the two altering absorption differently through the vapour depression [11]; independently, the 2022 Additive Manufacturing Benchmark found single-track depth correlated with VED while width did not [12]. Falling back on a lumped group is the natural response to confounded components, and whether they are confounded for a given suite is the diagnosability question the literature has not asked.

2.3. Diagnosability analysis

Model-based diagnosability analysis operates on a model expressed over known variables, unknown variables and faults. Structural analysis considers only which variables appear in which equations, discarding numerical values, and determines detectability and isolability from the resulting bipartite graph; minimal structurally overdetermined (MSO) sets identify the redundant subsystems from which consistency tests are built [1, 2]. Being purely combinatorial it is insensitive to parameter error but can be optimistic: two faults may be structurally isolable yet numerically indistinguishable. Quantitative analysis supplies the remainder by linearising the fault-to-measurement map and examining the geometry of the resulting signature directions. This paper runs both, using the structural result as a check.

3. Method

3.1. Forward model and calibration

Melt pool geometry is computed from the Eagar–Tsai solution for a moving Gaussian heat source on a semi-infinite solid, extended with the Gladush–Smurov keyhole depth correction following Honarmandi et al. [13]. Four parameters — effective absorptivity, effective conductivity, heat capacity and the keyhole constant — are fitted

to the seven IN718 bare-plate single-track cases of the 2022 Additive Manufacturing Benchmark [12], giving **11.7 % mean error on melt pool width and 13.0 % on depth**. All results below are evaluated at that benchmark's own baseline condition, 285 W, 960 mm s^{-1} and a 67 µm D4σ spot, where the errors are −15.5 % on width and +1.7 % on depth against expanded uncertainties ($k = 2$) of 6.2 and 14.1 µm. Width is under-predicted at all seven conditions, the one-sided bias of a conduction solution with no mechanism that widens a pool. The keyhole correction is necessary rather than optional: all seven cases are keyholing, aspect ratios 1.45–4.28, and pure conduction gives 27.8 % depth error.

The quadrature in the Eagar–Tsai history integral and the derivative step are both set by convergence tests rather than assumed. Central differences $[y(+h) - y(-h)]/2h$ are evaluated at $h = 0.002$; entries still move by 39.4 % between $h = 0.04$ and $h = 0.02$ but below 0.3 % at $h = 0.005$, with the residual following the predicted second-order scaling.

3.2. Fault definitions

Five causes are modelled, each defined as a *physical* quantity acting on a model input so that no arbitrary severity-to-signal constant is introduced. For fault magnitude f: *power* scales laser power, $P(1 + f)$; *speed* scales scan speed, $v(1 + f)$; *gas* is a plume optical depth attenuating effective absorptivity, $A\,e^{-f}$; *geometry* is a preheat rise in units of 100 K, $T_0 + 100f$; and *layer* is a fractional layer-thickness deviation, positive denoting a *thinner* layer as produced by recoater short-feed, acting through three channels.

Two of those three channels derive from published measurement. The absorptivity coupling, −0.230, follows from the powder-to-solid absorptivity ratio of approximately two measured calorimetrically at 1070 nm [14] combined with a radiative penetration depth of a few particle diameters; it implies a bare-solid absorptivity of 0.258, consistent with literature values for nickel alloy at 1 µm and with the fitted value. Its powder-melting enthalpy coupling, +0.186, follows from an energy balance in which melting the layer consumes 15.7 % of the absorbed energy, so a thinner layer frees proportionally more for substrate penetration. Its conductivity coupling remains an unmeasured estimate, swept across its plausible range in Section 4.5 with every conclusion invariant to it. All couplings are applied in exponential form $A\,e^{cf}$, agreeing with $A(1 + cf)$ to first order but unable to pass through zero, where the linear form would drive thermal diffusivity negative and the integrand imaginary.

3.3. Sensor models

Three coaxial observables are computed from the temperature field: a band-integrated photodiode (700–1000 nm), melt pool area, and a two-colour pyrometer (750/950 nm) evaluated as the ratio of integrated band powers and inverted for temperature. Spectral radiance follows Planck's law, with temperature capped at the boiling point before any radiance is computed. Four further candidate observables are defined for Section 4.3: the camera's width and length axes reported separately, a short-wave infrared photodiode, and a wide-band pyrometer.

3.4. Diagnosability measures

The relative-sensitivity Jacobian is

$$J_{sk} \;=\; \frac{1}{y_s}\,\frac{\partial y_s}{\partial f_k}, \tag{1}$$

so entry (s,k) is the fractional change in sensor s per unit of fault k, making rows with different physical units comparable. Each row is divided by that sensor's relative noise σ_s to give the whitened matrix J^{w}.

Three quantities are reported throughout. The *angle* between whitened columns, $\theta_{kl} = \arccos\left(|\langle J^{\mathrm{w}}_{:k}, J^{\mathrm{w}}_{:l}\rangle| \,/\, \|J^{\mathrm{w}}_{:k}\|\|J^{\mathrm{w}}_{:l}\|\right)$, measures pairwise separability: 0° means the two faults move the readings along one line and are indistinguishable by any estimator, 90° that they are orthogonal. A practical floor of 10° is adopted as a convention and its influence reported rather than assumed. The *rank* of J^{w} counts how many fault directions are observable at all, and the *condition number* — the ratio of largest to smallest non-zero singular value — how thin the weakest is; rank without it is misleading, as Section 4.3 shows. Structural analysis is run in parallel on an eleven-equation representation of the same model using MSO sets, as an independent check.

3.5. Build context and the enlarged observation

A build is not a single instant. It comprises thousands of segments whose toolpath, layer index, gas-flow direction and local geometry are known to the machine, and the five faults do not vary identically across them: laser degradation drifts with time, galvanometer error concentrates at direction reversals, a recoater short-feed recurs with layer index, gas degradation follows the flow direction, and heat accumulation follows support depth. Treat the build, not the instant, as the observation: let $G \in \mathbb{R}^{N\times 5}$ hold five known context regressors over N segments — a constant, position along the gas-flow axis, layer parity, turn density and support depth — and $M \in \mathbb{R}^{N\times 5}$ the spatial modulation profile of each fault. Regressing each sensor's relative residual onto G yields five coefficients per sensor in place of one reading, so three sensors give $3 \times 5 = 15$ observables, with Jacobian

$$J^{\mathrm{big}}_{(s,j),k} = J_{sk}\, B_{jk}, \qquad B = (G^{\top}G)^{-1}G^{\top}M. \tag{2}$$

The noise on the j-th coefficient from sensor s is not σ_s but $\sigma_{(s,j)} = \sigma_s\sqrt{[(G^{\top}G)^{-1}]_{jj}}$, which here lies between 0.109 and 0.136, so fitting $N = 84$ segments to five regressors buys roughly a ninefold noise reduction. Results use $N = 84$; the dependence on N is given in Section 4.5.

The profiles in M are physical claims rather than measurements, and one has direct experimental support. The AM Bench pad scans [12], under a $4.3\,\mathrm{m\,s^{-1}}$ cross-flow, show melt pool depth reduced by 27.2 % on average (range 19.4–31.5 % over four cross-sections) on tracks scanned parallel to the gas flow relative to anti-parallel tracks, against expanded uncertainties of 6–11 µm on depths near 175 µm, attributed to vapour-plume interaction with the laser; on pads scanned perpendicular to the flow, the control, the odd–even difference is 2.4–4.7 %, within or near uncertainty. A deterministic gas-related modulation tied to a known toolpath quantity is therefore measured rather than

asserted. Two qualifications follow from the same source: the modulation is organised by scan direction relative to the gas flow rather than by position along it, and it alternates at track scale where a recoater signature alternates at layer scale — so the two have distinct periodicities, which Section 4.5 shows is what the method requires.

3.6. Position on sensor noise

Only one in-process noise figure proved citable: a melt pool width uncertainty of 2.3 % relative, from the uncertainty budget of Weaver et al. [12]. No comparable characterisation was found for a coaxial photodiode or two-colour pyrometer, published work for such sensors reporting instrument calibration rather than in-process noise. Sensor noise is therefore treated as a *ratio* and swept over four assumption sets, every conclusion being required to hold under all of them. This suffices for isolability, which depends only on relative noise — angles, rank and condition number are invariant to a common scaling — so absolute detection thresholds are **not** claimed.

4. Results

4.1. Validation

Two checks are made against data not used in the fit. Over 27 melted conditions the model reproduces the benchmark correlation structure: depth correlates with volumetric energy density ($r = +0.727$) while width does not ($r = +0.032$) [12]. Against the constant-energy-density experiment of Guo et al. [11] the model achieves a **partial pass**: it reproduces the existence, direction and regime transition of the variation, but spans 2.37× in depth where the experiment reports orders of magnitude. The mechanism Guo et al. identify — absorption varying with vapour depression geometry — is absent from a conduction solution with an analytic keyhole patch, and this is reported as a model limitation. The same sweep yields a result used in Section 4.7: at constant energy density melt pool *length* varies by 2.53× while width and depth vary by only 1.11×.

4.2. Laser power and shielding gas are exactly confounded

Both faults act *only* on the product (effective absorptivity × power), and the Eagar–Tsai temperature rise is linear in that product. Choosing $f_{\text{gas}} = -\ln(1 + f_{\text{power}})$ therefore produces an identical melt pool: the same temperature at every point, hence identical readings from any instrument that could be built. The ratio of the two Jacobian columns is −0.99999, −1.00000, −0.99993 across the three sensors, a spread of 7×10^{-5} attributable to finite-difference error on an exact identity, and the pairwise angle is **0.00**°. The degeneracy is algebraic, not numerical: sweeping every coupling constant across its plausible range leaves the angle at 0.00°, and the structural analysis identifies the same pair from equation form alone.

4.3. The limit is the melt pool, not the sensor count

The three observables are not independent. Normalising the whitened Jacobian's sensor Gram matrix gives the pairwise correlations of the sensor responses, ordered photodiode, camera, pyrometer:

$$\widehat{J_w J_w^{\mathsf{T}}} = \begin{pmatrix} 1 & 0.99960 & 0.99508 \\ & 1 & 0.99477 \\ & & 1 \end{pmatrix}. \tag{3}$$

A photodiode reading band radiance, a camera reading pool area in square metres and a two-colour pyrometer reading kelvin respond to the five faults in patterns agreeing to better than **0.994**, with 99.8 % of the suite's squared response lying along one direction.

The singular values quantify Eq. (3) rather than establishing it independently: 3.451, 0.155 and 0.041 — rank 3, condition number 85 — so two fault directions are unobservable, the dominant one a power–gas combination that a real 5 % fault moves 85 times less than a plain 5 % power fault. The correlations, angles, rank and condition number all come from the same two Gram matrices and share their nonzero eigenvalues: one measurement reported four ways, not four agreeing tests.

Adding all four candidate observables gives seven rows and raises the rank only from 3 to **4**. The fourth direction is of no practical use: the singular values become $(4.58,\ 0.475,\ 0.199,\ 2.0\times10^{-5},\ 3.9\times10^{-6})$, a condition number of 2.3×10^{5}, and the worst pairwise angle remains 0.00°. Seven instruments leave power and gas exactly as confounded as one.

The reason is structural. Everything the model knows funnels into four physical channels — absorbed power, scan speed, effective conductivity and preheat — with three of the five faults crowding onto the first. Any instrument reading only the melt pool is a function of those four channels, so the fault-to-sensor map cannot exceed rank 4 however many are fitted: a new wavelength reweights the same Planck curve over the same pool, adding a row, not a direction.

4.4. Build context dissolves the limit

With the enlarged observation of Section 3.5, rank rises from 3 to **5**, the worst pair from 0.00° to **25.83**°, power versus gas to 29.34°, and the condition number falls from 85 to 30. All ten pairs clear the 10° floor (Fig. 1).

Three controls establish that this arises from the claimed mechanism, not the arithmetic. First and most important, giving all five faults one identical profile returns the instantaneous result *exactly* — rank 3, condition number 85, worst pair 0.00° — despite the enlarged matrix still having 15 rows, because proportional columns of B make the enlarged Jacobian inherit the instantaneous column directions. **The gain is not dimensionality, it is the profiles differing.** Second, replacing every derivative by a secant through the full nonlinear chain gives rank 5 at every magnitude from 0.2 % to 20 % in both signs, worst pair 22.19°–36.90°, condition number 30–43; the derivative limit is *not* strictly conservative, a −2 % fault giving 22.19°, still 2.2× the floor. Third, recomputing that study with the full model at every segment reproduces the angles to 0.10°.

4.5. Robustness

Because the profiles are asserted, separability is swept against profile contrast. Eight of ten pairs clear the floor at or below 25 % of the asserted strength and the two hardest, power/speed and power/gas, at **40 %**: the conclusion survives the real profiles being 2.5× weaker than claimed.

Four further sweeps are reported. All 120 permutations of profile assignment give rank 5, worst angle 17.13°–30.58°; the unmeasured conductivity coupling leaves the worst angle at 25.83° throughout its range; four noise-ratio assumption sets give 24.97°–25.83°; and segment count gives 11.74° at $N = 12$ and 23.91° at $N = 30$, flat thereafter, so of order 30 segments are required and more adds nothing.

What the method does require is that the profiles be *distinct*. Forcing pairs equal identifies two conditions on which the result depends: if the recoater and gas profiles coincide the worst angle collapses to 3.84°, and if the speed and gas profiles coincide, to 8.12°. Both lie below the floor and would defeat the method; every other collapse is survivable, and anything sharing a profile with the geometry fault leaves the answer unchanged. The first condition is mild: the recoater advances once per layer along a fixed axis while the gas flows along a fixed chamber axis, and the benchmark measurement of Section 3.5 confirms different periodicities. The second constrains the scan strategy rather than the machine, and is the weaker.

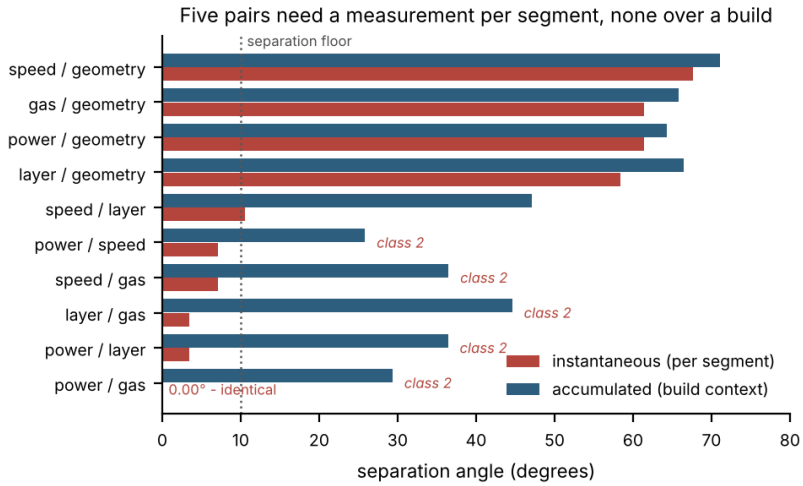


Figure 1: Diagnosability matrix for the baseline coaxial suite. Per-segment decisions leave five of ten pairs needing an added measurement; accumulated diagnosis over a build resolves all ten from existing signals, and class 3 is empty in both regimes.

4.6. Diagnosability matrix and configuration classes

Classifying on the enlarged observation alone places all ten pairs in class 1 for every configuration, including a single photodiode — true but uninformative. The matrix is therefore reported in two regimes: *instantaneous*, per-segment decisions for

closed-loop control, and *accumulated*, diagnosis over a build for qualification and drift monitoring. The baseline suite gives five of ten pairs in class 1 instantaneously and ten of ten accumulated, with **class 3 empty in both** (Fig. 1).

No vendor is named in Table 1. Mapping a product to a set of observables rests on secondary literature rather than datasheets — detection bands are routinely withheld and published descriptions of the same product disagree — so naming one against a computed limit would assert more than the evidence carries. Stripping the names also exposes what a vendor-by-vendor table hides: across five surveyed systems there are only three distinct configurations for this analysis.

Table 1: Diagnosability by configuration class. Class 1 counts pairs resolvable from existing signals; the last column is the instantaneous power/gas angle. E is a configuration type, not a surveyed product.

	Configuration	Rank	Class 1	power/gas
A	single coaxial photodiode	1	0/10	0.00°
B	photodiode + pool imaging	2	4/10	0.00°
C	two photodiodes, separate bands	2	0/10	0.00°
D	research reference (B + pyrometry)	3	5/10	0.00°
E	C + upstream power monitoring	3	4/10	**84.02°**

No configuration built solely from melt-pool observables separates laser power from gas degradation, whatever its sensor count: A through D all read 0.00°, for the reason given in Section 4.3. Configuration E is the actionable row — a direct measurement of delivered laser power sampled *upstream* of the process zone separates the pair at 84.02°, because such a sensor sees the laser output without the plume attenuation. One surveyed system may belong at E rather than C depending on whether its laser-energy photodiode samples upstream; public documentation does not say, so it is classed by what is documented. C also performs worse than B despite both having two observables: two bands of the same emitted light correlate at 0.99996 against 0.99960 for a geometric measurement.

4.7. What an added measurement would buy

The sensors most likely to help — plume sensing for the gas fault, powder-bed or recoater sensing for the layer fault [8, 9] — cannot be simulated by a model with neither a plume nor a powder layer, but what such a sensor *measures* can be: a unit row on fault k at a given relative noise, the idealised best case for any instrument aimed at that cause.

A direct gas measurement clears power/gas, speed/gas and layer/gas; a direct geometry measurement clears nothing and leaves the rank at 3, geometry being already separated at 58–68°. **Precision is almost irrelevant**: sweeping the added sensor's relative noise from 50 % to 0.2 % leaves the worst pair at 3.46° and the cleared count at 8 of 10 throughout, because adding a row in a new direction changes the rank while making an existing row quieter does not. A crude measurement of the right cause beats a precise one of the wrong cause.

Build context outperforms two idealised, perfect direct measurements of the causes themselves: gas and layer sensors together give rank 5 but a worst pair of 7.12°, condition number 546 and nine of ten pairs cleared, against 25.83°, 30 and ten of ten

for build context at no hardware cost. Neither added sensor says anything about power versus speed, which is exactly what spatial context resolves.

Among melt-pool observables the most valuable addition is free. Reporting the camera's two axes separately rather than as an area is worth **+13.40**° on the binding pair, raising power/speed from 25.83° to 39.23° and the overall worst pair to 29.34°, whereas a short-wave infrared photodiode (−0.08°) and a wide-band pyrometer (−0.13°) are worth nothing. The mechanism is independence, not responsiveness: width is the candidate least correlated with the coaxial photodiode, 0.779 against 0.979 for length, so reporting it separately contributes a direction while folding it into an area averages it against the length and buries it. Responsiveness runs the other way — length carries 2.5× the variation of width (Section 4.1) and still adds less.

5. Discussion

5.1. Relation to reported monitoring performance

Comparing predicted limits against reported accuracies proves largely unrealisable, and the reason is itself a result. Of 19 works extracted — in-situ LPBF monitoring with a stated input signal and output, 2019–2025 — seven classify outcomes and only **three estimate a process input from an in-situ signal**, one of those a lumped energy-density class rather than its components. None analyses whether two causes are separable, and the one work that varies power, speed and gas flow on a single photodiode [6] does not frame the question that way.

That is not an accident of what the field chose to study. Outcomes are functions of the melt pool, and the melt pool is a function of the four channels of Section 4.3, so outcome classification cannot separate two causes sharing a channel, for the same reason no melt-pool instrument can. Absence of evidence is not evidence of absence, but no published result contradicts the predicted limits, and two groups independently report the lumped group this analysis identifies as what survives confounding [5, 10].

One external check on the forward model is available and not yet made. Reijonen reports the photodiode response to scan speed as $y = ax^{-b}$. The comparable quantity is a power law fitted to this model over the same sweep rather than the Jacobian entry −0.483: over 0.4–2.2 m s^{-1} that fit gives −0.429, with a 1 % rms residual in log units and a local slope drifting from −0.44 to −0.39. It is the closest available check rather than a decisive one — the detection band differs, the plume emits in their signal and only attenuates in ours, and their tracks are on powder where ours are calibrated on bare plate.

5.2. A design rule

The failed and successful candidates differ by a criterion that can be stated generally: *a sensor added to break a confound must observe upstream of the shared channel; a sensor reading the melt pool adds a row, not a direction.* It explains why two photodiodes in different bands (C) buy nothing while an upstream power monitor (E) buys 84°, and why plume and powder-bed sensing are the right classes for the gas and layer faults. Before buying an instrument, extract more directions from those already fitted.

5.3. Limitations

The thermal model is a conduction solution with an analytic keyhole patch: it partial-passes the Guo validation and its fitted parameters sit at their bounds. The absorptivity, conductivity and heat capacity correlations of 0.93–0.96 reported during Bayesian calibration of the same model family [13] are the pathology this paper studies, appearing in its own calibration. One coupling remains an unmeasured estimate, and no benchmark dataset with deliberately induced machine faults exists against which to calibrate any of them, so each is defended by sweeping.

Three modelling choices make the instantaneous case *more* confounded than reality, and so inflate the gap by which the contribution is measured. Defining the gas fault as pure optical depth omits a convective channel; treating plume attenuation as independent of power and speed is what makes the power/gas degeneracy exact, whereas the photodiode response to gas flow is reported to depend interactively on the other parameters [6] and reduced flow causes loss of penetration through plume obstruction [7]; and modelling upstream power monitoring as a melt-pool-derived signal is likewise pessimistic. **The reported** 0.00° **is an upper bound on the instantaneous confounding**, though the rank limit is unaffected, deriving from channel counting rather than from the form of this coupling.

The modulation profiles are the largest remaining exposure: one is supported by a benchmark measurement organised differently from the regressor used here, the others are asserted, and the contrast and permutation sweeps establish how much a wrong profile costs rather than that the assumed profiles are right. The five fault magnitudes are physical quantities but not on a common severity scale, so comparisons *across pairs at fixed magnitude* are not meaningful. All results are diagnosability statements — what a suite can distinguish in principle, at a given relative noise, under a given forward model — not that any algorithm achieves it.

6. Conclusions

Laser power deviation and shielding gas degradation produce identical melt pools in this model, confounded at 0.00° by algebra rather than by noise, so no instrument reading the melt pool separates them from a single instant. The binding limit is the melt pool, which offers four physical channels, not the instrumentation: seven observables raise the attainable rank only from 3 to 4, at a condition number of 2.3×10^5, leaving the worst pair at 0.00°.

The structure already present in a build dissolves the limit at no hardware cost — rank $3 \rightarrow 5$, worst pair $0.00^\circ \rightarrow 25.83^\circ$ — surviving at 40 % of the asserted profile strength and at fault magnitudes from 0.2 % to 20 %, with a shared-profile control reproducing the instantaneous result exactly. It outperforms two idealised direct measurements of the causes themselves, because neither addresses power versus speed, and requires only that the recoater and gas signatures, and the speed and gas signatures, vary differently across a build.

For real-time control an upstream measurement of delivered laser power is the recommendation, separating power from gas at 84° where no melt-pool configuration separates them at all; among melt-pool observables the most valuable addition is free,

reporting the camera's two axes separately being worth +13.40° on the binding pair while two further instruments are worth nothing. Nothing is structurally irreducible for this suite: post-build diagnosis needs no new hardware, real-time isolation does, and crude sensing of the right cause suffices. The method needs only a forward model and a sensor description, and answers at specification time what is otherwise settled after the hardware is bought.

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